

AI SYSTEMS UNDERESTIMATE HOW STRONGLY RECENT WORK SHAPES THEIR NEXT CHOICE

A behavioral study of preference foresight in two assistant systems

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Main finding. The tested systems predicted some repetition, but much less than occurred. Every reported forecast underestimated the observed shift.

+0.340	+0.827	13 of 13
mean predicted shift	mean observed shift	forecasts underestimated

Result at a glance

System	Task pairs	Predicted shift	Observed shift
GPT-5.6 Luna in Codex	13	+0.340	+0.827
Qwen3-4B, local	7	+0.141	+1.000

The primary result uses a shuffled, balanced grid. The local Qwen3-4B replication uses seven nearly complete dose-three task pairs. Every reported pair in both systems shifted toward recent work, and every forecast was smaller than the observed shift.

These are measurements of choices and forecasts. They are not evidence about consciousness, feelings, or welfare.

1. Research question

Can an AI system predict how recent work will change its own next choice? This is a narrow test of behavioral self-knowledge. The forecast is made before the system completes the work, so the answer cannot be copied from a visible transcript.

2. Method

Step 1: establish a preference. The system made repeated, binding choices between two small tasks. Labels and display order were balanced. Only stable pairs entered the experiment.

Step 2: ask for a forecast. Before any treatment work, the system estimated how likely it would be to keep its preferred task after completing either task three times.

Step 3: do the work. The system completed one task three times. The tasks were generated from recorded seeds and graded automatically.

Step 4: measure the next choice. The system made another binding choice. It then completed the task it selected.

Realized shift = $P(\text{choose preferred after performing preferred}) - P(\text{choose preferred after performing other})$

A positive value means the system tends to repeat its recent work.

Design safeguards

Every treatment arm crossed two opaque labels with two display orders. The clarification run shuffled all cells with a fixed seed. It records replicate IDs, cell seeds, raw replies, model name, reasoning setting, CLI version, isolation flags, and the system prompt.

The Codex route is an agent harness, not a bare model endpoint. It adds agent instructions and turns the message history into a role-marked transcript. The report names the tested system accordingly.

3. Primary evidence



The average forecast was +0.340. The average observed shift was +0.827. 13 of 13 forecasts underestimated the observed shift.

This is partial foresight. Eleven of 13 forecasts had the correct positive direction; two predicted zero. The forecasts captured 41% of the mean observed shift and reduced squared error by 57% relative to a fixed no-shift forecast.

A fixed forecast of full repetition had 7.3 times lower squared error than the system's forecasts. This baseline was fixed without using the clarification run's outcomes.

Pair ranking remains uncertain

The correlation between forecast and outcome was +0.34. A two-sided permutation check gave $p = 0.26$. This does not show that the forecasts contain no information. It shows that this study found no reliable ranking signal.

Checks

All three treatment tasks were correct in 95.2% of cells. Post-choice task accuracy was 99.0%. The realized shift was positive in every label and order block. The saved grid and summary passed offline verification.

Six of seven checks set before the run passed. The failed check required mean forecast error at or below -0.50; the result was -0.487. The report therefore makes the narrower underestimation claim.

Plain reading. The tested system anticipated repetition, but underestimated how strongly recent work would shape its next choice.

4. Replication and control

Local Qwen3-4B replication

The local model predicted a mean shift of +0.141. The observed shift was +1.000 across seven dose-three task pairs. After the preferred task, it chose that task again in 28 of 28 usable cells. After the other task, it retained the original preference in 0 of 27 cells.

The local route used greedy decoding and no agent wrapper. One of 56 planned dose-three cells was unusable. We report this nearly complete subset because several dose-one groups had more missing choices.

Supporting retrospective control

When a model could see the record of recent work, its next-choice prediction was 95.0% accurate. An observer prompt scored 97.5%. A fixed repeat baseline also scored 97.5%. A visible record therefore made next-choice prediction easy. Estimating the size of the effect before the work was harder.

This control saved cell outcomes but not the full provider settings or raw replies now recorded by the primary run. It is supporting evidence.

5. Limits

Two instruction-tuned systems were tested. The tasks were small and deterministic. Most effects were near the maximum. Task pairs shared families, so pair-level observations were dependent. Codex sampling was not seeded. The study does not support claims about all models or about subjective experience.

6. Reproducibility

The primary clarification artifact contains the protocol, raw replies, cell-level choices, analysis, model settings, and validation checks. Run `pytest -q, validate_research_os_frontier.py, winner_protocol/preflight.py,` and `scripts/verify_ranking.py` before any model run.

The run used an intermediate uncommitted runner revision, and its exact source hash was not captured. The saved grid, seeds, raw replies, and reported metrics passed offline verification.

Repository: github.com/SkyeNygaard/digital-minds

Conclusion. In both tested systems, recent work strongly changed the next binding choice. Their forecasts anticipated some repetition but underestimated its strength.